

N100 Financial Intelligence

Analyst Guide

Streamlit dashboard • Financial screening • Peer and sector analysis • FastAPI • Tearsheets • Annual reports

Project	N100 Financial Intelligence
Universe	NIFTY 100 companies
Dashboard	Streamlit
API	FastAPI 1.0.0
Database	SQLite

Purpose

A practical guide for analysts, mentors and reviewers using the project for structured financial research.

1. Analyst Operating Model

The system combines financial data, analytical modules, a Streamlit interface and a FastAPI service. Move from broad discovery to focused company research.

Recommended workflow

- **Home:** understand the NIFTY 100 universe, sector distribution and top composite scores.
- **Screener:** narrow the universe using financial conditions.
- **Company Profile:** inspect indicators, historical data and Pros & Cons.
- **Peer Analysis:** compare a company with its peer group.
- **Trend Analysis:** review historical growth and data coverage.
- **Sector Analysis:** understand sector-level context and constituents.
- **Capital Allocation Map:** investigate cash-flow and allocation patterns.
- **Annual Reports:** retrieve primary-source documents.
- **API:** automate or reproduce data retrieval.

Core data objects

Object	Use
Companies	Company identity and profile information
Profit & Loss	Sales, expenses, operating profit, net profit and EPS history
Balance Sheet	Assets, liabilities, reserves and borrowings
Cash Flow	Operating, investing, financing and net cash flow
Financial Ratios	Profitability, leverage, cash-flow quality and CAGR metrics
Market Cap	Market capitalization and valuation history
Peer Groups / Percentiles	Relative peer membership and KPI ranking
Sectors	Broad sector, sub-sector and index-weight context
Documents	Annual-report links by company and year

Use the dashboard for exploration and the API for repeatable retrieval. Important conclusions should be traced to historical data or primary-source reports.

2. Startup and Environment

Run commands from the repository root. The verified working environment for this project uses the Anaconda Python installation.

Start Streamlit

```
streamlit run src/dashboard/app.py
```

Start FastAPI

```
& "$env:USERPROFILE\anaconda3\python.exe" -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

Useful URLs

URL	Purpose
http://127.0.0.1:8000/	Basic API information
http://127.0.0.1:8000/api/v1/health	Health and database row-count check
http://127.0.0.1:8000/docs	Interactive Swagger documentation
http://127.0.0.1:8000/redoc	Alternative API documentation

Health check

```
Invoke-RestMethod http://127.0.0.1:8000/api/v1/health | ConvertTo-Json -Depth 5
```

- A healthy response contains status = ok, database row counts, uptime and version.
- Keep the API terminal running while testing from a second terminal.
- If the default python environment causes Uvicorn to exit, use the explicit Anaconda Python path.

3. Dashboard: Home and Company Profile

Home — 01_home.py

The Home screen is titled **Nifty 100 Financial Intelligence** and provides a financial overview of the universe.

- **NIFTY 100 Sector Distribution:** understand the broad sector mix.
- **Top 5 Composite Scores:** identify candidates for deeper research.

Company Profile — 02_profile.py

The Company Profile screen is titled **Company Profile** and provides a detailed view for a selected company.

- **Company header:** confirms the selected company.
- **Key Financial Indicators:** review the principal displayed metrics.
- **Historical sections:** inspect changes over time.
- **Pros & Cons:** review the generated qualitative summary.

Analyst questions

- Is profitability improving or deteriorating?
- Does growth translate into cash generation?
- Is leverage becoming more or less demanding?
- Do the quantitative trends support the Pros & Cons?

4. Dashboard: Financial Screener

The Financial Screener narrows the NIFTY 100 universe using optional financial filters.

Filter	Meaning
Minimum ROE	Minimum return on equity
Maximum Debt-to-Equity	Upper leverage limit
Minimum FCF	Minimum free cash flow
Sector	Restrict to a broad sector
Minimum 5-year Revenue CAGR	Minimum historical revenue growth
Minimum 5-year PAT CAGR	Minimum historical profit growth
Maximum P/E	Upper valuation threshold

Practical workflow

- Start with a small number of filters and observe the result set.
- Add leverage or valuation constraints when risk or price discipline matters.
- Use sector filtering for like-for-like investigation.
- Validate shortlisted companies with Profile, Peers, Trends and reports.

API equivalent

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1&max_pe=30"
```

The API validates non-negative thresholds for ROE, debt-to-equity, FCF, revenue CAGR and PAT CAGR; maximum P/E must be greater than zero.

A screen is a selection mechanism, not a final recommendation. Passing a screen should trigger deeper investigation.

5. Dashboard: Peer and Trend Analysis

Peer Analysis — 04_peers.py

- **Peer group selection:** identifies the comparison set.
- **Company vs Peer Average:** highlights relative strengths and weaknesses.
- **Peer KPI Comparison:** supports metric-level benchmarking.

API example

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS/peers/compare"
```

Trend Analysis — 05_trends.py

- **Year-over-Year Growth:** inspect annual changes.
- **Growth Summary:** identify sustained or changing trends.
- **Data Coverage:** check available historical depth.

Analyst discipline

- Separate sustained trends from one-year spikes.
- Confirm the history is long enough for the question being asked.
- Investigate sharp metric changes through the underlying statements.

6. Dashboard: Sector and Capital Allocation

Sector Analysis — 06_sectors.py

- **Revenue Growth vs ROE:** examine growth and return relationships.
- **Sector Median KPIs:** compare companies with sector context.
- **Companies in Selected Sector:** inspect sector constituents.

API examples

```
curl "http://127.0.0.1:8000/api/v1/sectors"  
curl "http://127.0.0.1:8000/api/v1/sectors/Information%20Technology/companies"
```

Capital Allocation Map — 07_capital.py

- Review the displayed capital-allocation views and supporting summaries.
- Use cash-flow history to distinguish internally generated cash from financing activity.
- Compare allocation patterns with profitability, leverage and growth.

Recommended conclusion format

What happened → period → supporting metric → implication/risk → primary source to check next.

7. Annual Reports and Company Tearsheets

Annual Reports — 08_reports.py

The Annual Reports screen provides an annual-report repository for available company and year combinations.

- Select the company of interest.
- Review the available report years.
- Open the report when a metric needs primary-source validation or more business context.

Tearsheet API

```
curl -o TCS_tearsheet.pdf http://127.0.0.1:8000/api/v1/companies/TCS/tearsheet
```

The tearsheet endpoint returns the pre-generated company PDF; a missing output returns 404.

Company data endpoints

Endpoint	Use
/api/v1/companies	List the NIFTY 100 companies
/api/v1/companies/{id}	Company profile
/api/v1/companies/{id}/pl	Historical P&L
/api/v1/companies/{id}/bs	Historical balance sheet
/api/v1/companies/{id}/cashflow	Historical cash flow
/api/v1/companies/{id}/ratios	Historical financial ratios
/api/v1/companies/{id}/tearsheet	Company PDF

For material claims, cross-check the underlying historical data and annual report.

8. FastAPI: Analytics Endpoints

The FastAPI application is version 1.0.0. The analytics router exposes screening, sector, peer, market-cap, portfolio and document retrieval.

Endpoint	Purpose
/api/v1/screener	Apply optional financial filters
/api/v1/sectors	Return broad sectors and company counts
/api/v1/sectors/{sector}/companies	Return companies in a sector
/api/v1/peers/{group_name}	Return peer members and percentile metrics
/api/v1/companies/{id}/peers/compare	Return peer radar-style KPI data
/api/v1/market-cap/{id}	Return market-cap and valuation history
/api/v1/portfolio/stats	Return generated portfolio statistics
/api/v1/companies/{id}/documents	Return annual-report links

Common examples

```
curl http://127.0.0.1:8000/api/v1/companies
curl http://127.0.0.1:8000/api/v1/companies/TCS
curl http://127.0.0.1:8000/api/v1/companies/TCS/pl
curl http://127.0.0.1:8000/api/v1/companies/TCS/bs
curl http://127.0.0.1:8000/api/v1/companies/TCS/cashflow
curl http://127.0.0.1:8000/api/v1/companies/TCS/ratios
```

PowerShell

```
Invoke-RestMethod http://127.0.0.1:8000/api/v1/companies/TCS/ratios | ConvertTo-Json -Depth 6
```

Open **/docs** while FastAPI is running to inspect and try the registered routes.

9. API Responses and Data Interpretation

Expected response statuses

Status	Typical meaning
200	Request completed successfully
400	Invalid screener parameter or other client input
404	Company, sector, peer group, document or tearsheet not found

Invalid-input examples

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=-5"  
curl "http://127.0.0.1:8000/api/v1/companies/NOTREAL"  
curl "http://127.0.0.1:8000/api/v1/sectors/NotARealSector/companies"
```

Before interpreting a result

- Confirm the company identifier.
- Check historical coverage for the company.
- Check that the metric exists for the required period.
- Compare ratios with the corresponding statements for unexpected values.
- Use peer and sector context before labeling a value strong or weak.

Coverage note

The current database contains 92 companies, but historical coverage is not identical across all companies and statements. Long-horizon comparisons should therefore state the available period.

10. Troubleshooting

Uvicorn starts and immediately shuts down

Use the explicit Anaconda Python executable:

```
& "$env:USERPROFILE\anaconda3\python.exe" -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

Leave that terminal running and test from a second terminal.

Health endpoint says connection refused

- Confirm Uvicorn is still running and has not printed "Shutting down".
- Confirm port 8000 is being used.
- Use `/api/v1/health` rather than an outdated health path.

Company endpoint returns 404

- Check the company identifier and use `/api/v1/companies` to retrieve the supported universe.

Sector endpoint returns 404

- Use `/api/v1/sectors` to obtain available broad-sector names, then use the returned spelling.

Tearsheet endpoint returns 404

- The requested pre-generated PDF may not exist; use Annual Reports and the documents endpoint for source documents.

pytest command is unavailable

```
& "$env:USERPROFILE\anaconda3\python.exe" -m pytest -q
```

Use the verified Anaconda environment for the project test suite.

11. Analyst Quick Checklist

Before publishing or relying on an analysis

Check	Done
Company identifier confirmed	■
Historical coverage reviewed	■
P&L trend reviewed	■
Balance-sheet / leverage trend reviewed	■
Cash-flow trend reviewed	■
Key ratios reviewed	■
Peer comparison reviewed	■
Sector context reviewed	■
Valuation context reviewed where relevant	■
Pros & Cons checked against quantitative evidence	■
Annual report checked for material claims	■
Tearsheet/supporting artifact saved if required	■

API checklist

- Health endpoint returns status: ok.
- Company list returns the expected universe.
- Historical company endpoints return data.
- Screener accepts valid filters and rejects invalid thresholds.
- Sector and peer endpoints return expected groups.

Research note template

Company:
Period reviewed:
Key observation:
Supporting KPI(s):
Peer/sector context:
Primary-source document checked:
Risk / limitation:
Conclusion:

12. Complete API Reference

#	Method	Endpoint
1	GET	/api/v1/health
2	GET	/api/v1/companies
3	GET	/api/v1/companies/{company_id}
4	GET	/api/v1/companies/{company_id}/pl
5	GET	/api/v1/companies/{company_id}/bs
6	GET	/api/v1/companies/{company_id}/cashflow
7	GET	/api/v1/companies/{company_id}/ratios
8	GET	/api/v1/companies/{company_id}/tearsheet
9	GET	/api/v1/screener
10	GET	/api/v1/sectors
11	GET	/api/v1/sectors/{sector}/companies
12	GET	/api/v1/peers/{group_name}
13	GET	/api/v1/companies/{company_id}/peers/compare
14	GET	/api/v1/market-cap/{company_id}
15	GET	/api/v1/portfolio/stats
16	GET	/api/v1/companies/{company_id}/documents

Dashboard screen reference

File	Screen
01_home.py	Nifty 100 Financial Intelligence
02_profile.py	Company Profile
03_screener.py	Financial Screener
04_peers.py	Peer Analysis
05_trends.py	Trend Analysis
06_sectors.py	Sector Analysis
07_capital.py	Capital Allocation Map
08_reports.py	Annual Reports

The application is intended to support structured financial research: screening → historical analysis → peer/sector context → cash-flow/capital-allocation review → primary-source validation.